

Your grade: 100%

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To pass you need at least 80%. We keep your highest score.

Next item →

1.

What is the principle upon which Monte-Carlo integration is developed?

1 / 1 point
- ☒ We write the integral as an expectation, then approximate the expectation with a sample mean based on sampling from a known pdf related to the integrand.

☐ We write the integral as an expectation, then approximate the expectation with a sample standard deviation based on sampling from a known pdf related to the integrand.

☐ We sample the integrand on a regular grid and use the trapezoidal rule to approximate the integral.

☐ We sample the integrand on a regular grid and use the rectangular rule to approximate the integral.

☒ Correct
Yes, that is correct.
2.

When is it necessary to use an importance density when we are implementing Monte-Carlo integration?

1 / 1 point
- ☒ When we cannot draw random samples from the pdf $f(x_{k-1} \mid \mathcal{Z}_{k-1})$.

☐ When the pdf $f(x_{k-1} \mid \mathcal{Z}_{k-1})$ is either a uniform or Gaussian distribution.

☐ When the state x_k has more than a single element.

☐ When we cannot draw random samples from the pdf $f(x_k \mid \mathcal{Z}_{k-1})$.

☒ Correct
Yes, that is correct.
3.

What is the purpose of weight normalization?

1 / 1 point
- ☐ Weight normalization improves the accuracy of the particle filter.

☐ Weight normalization is required whenever the state x_k has more than one element.

☐ We must always use weight normalization if we are using the Monte-Carlo method with an importance density.

☒ It allows us to skip computing $f(z_k \mid \mathcal{Z}_{k-1})$ in the update step.

☒ Correct
Yes. This is correct.
4.

Execute the Coursera Lab Jupyter notebook for Lesson 4.2.3, which implements the Monte-Carlo method to approximate an integral with and without importance functions.

1 / 1 point
- Use the default `nx = 0.75`. Also make sure that the code executes `randn("state",0)` and `rand("state",1)` before generating random samples; otherwise the result of running the code will not agree with the expectations of the quiz auto-grader.

Set `N = 15000`; then, run the code. What is the root-mean-squared error between the true integral value and the estimate produced by the Monte-Carlo method over the final 2000 iterations? Enter your answer with four digits to the right of the decimal point.

0.0006

☒ Correct
Yes. That is correct.
5.

Execute the Coursera Lab Jupyter notebook for Lesson 4.2.3, which implements the Monte-Carlo method to approximate an integral with and without importance functions.

1 / 1 point
- Use the default `nx = 0.75`. Also make sure that the code executes `randn("state",0)` and `rand("state",1)` before generating random samples; otherwise the result of running the code will not agree with the expectations of the quiz auto-grader.

Set `N = 15000`; then, run the code. What is the root-mean-squared error between the true integral value and the estimate produced by the (Monte-Carlo with) importance-density method over the final 2000 iterations? Enter your answer with four digits to the right of the decimal point.

0.0034

☒ Correct
Yes. That is correct.
6.

Execute the Coursera Lab Jupyter notebook for Lesson 4.2.3, which implements the Monte-Carlo method to approximate an integral with and without importance functions, and with weight normalization.

1 / 1 point
- Use the default `nx = 0.75`. Also make sure that the code executes `randn("state",0)` and `rand("state",1)` before generating random samples; otherwise the result of running the code will not agree with the expectations of the quiz auto-grader.

Set `N = 15000`; then, run the code. What is the root-mean-squared error between the true integral value and the estimate produced by the (Monte-Carlo with importance-density and) weight-normalization method over the final 2000 iterations? Enter your answer with four digits to the right of the decimal point.

0.0026

☒ Correct
Yes. That is correct.
7.

For vector variable x having two elements, what is:

1 / 1 point
- $$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x^T x + 3) \delta(x - x_0) dx_1 dx_2,$$

if:

$$x_0 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}?$$

7

☒ Correct
Yes, that is correct.
8.

Suppose that $g(x) = x$ and $f(x) = \frac{1}{2} \sum_{i=1}^3 \delta(x - x^{(i)})$, where $x^{(1)} = 0.3$, $x^{(2)} = 0.5$, and $x^{(3)} = 1$. What is the value of $\hat{\mu} = \int_{-\infty}^{\infty} g(x) f(x) dx$? (Enter your answer with two digits to the right of the decimal place.)

1 / 1 point
- 0.6

☒ Correct
Yes, that is correct.
9.

Execute the Coursera Lab Jupyter notebook for Lesson 4.2.5, which implements a method of visualizing a pdf using kernel functions.

1 / 1 point
- Use the default `nSamples = 500` and `sigma = 1`. Also make sure that the code executes `randn("state",0)` and `rand("state",0)` and `randg("state",0)` before generating random samples; otherwise the result of running the code will not agree with the expectations of the quiz auto-grader.

Set `width = 0.6` and change the kernel function to the tri-weight formula; then, run the code. What is the Hellinger distance between the approximated pdf and the true pdf? Enter your answer with four digits to the right of the decimal point.

0.0120

☒ Correct
Yes, that is correct.
10.

Execute the Coursera Lab Jupyter notebook for Lesson 4.2.5, which implements a method of visualizing a pdf using kernel functions.

1 / 1 point
- Use the default `sigma = 1`. Also make sure that the code executes `randn("state",0)` and `rand("state",0)` and `randg("state",0)` before generating random samples; otherwise the result of running the code will not agree with the expectations of the quiz auto-grader.

Set `width = 0.6` and `nSamples = 2000` and change the kernel function to the tri-weight formula; then, run the code. What is the Hellinger distance between the approximated pdf and the true pdf? Enter your answer with four digits to the right of the decimal point.

0.0024

☒ Correct
Yes, that is correct.